CSI3005 ADVANCED DATA VISUALIZATION TECHNIQUES • SLOT E1 + TE1

CityPulse 360°: Vellore Urban Intelligence & R Architecture

Case Study Assessment Report, Visual Encoding Justification, Critique, Streaming Adaptation & Complete Spatial Engine

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Location: Vellore, Tamil Nadu

CSI3005 THEORY ASSESSMENT OVERVIEW

Report Structure & 10-Mark Rubric Alignment

TASK 01 • 2 MARKS

Frame the Problem

Audience, decision goal, dataset schema, and two analytical questions.

TASK 02 • 3 MARKS

Choose & Defend

Two visual encodings, labelled sketches, and alternative comparisons.

TASK 03 • 3 MARKS

Critique & Correct

Deliberately flawed sketch, two design errors, and corrected chart.

TASK 04 • 2 MARKS

Streaming Data

Continuous telemetry window, update cadence, and aggregation trade-off.

CODEBASE AUDIT • CITYPULSE360.R

System Scale & Data Architecture Metrics

1,008

Synthetic Records

Seeded via set.seed(360)

8

Vellore Urban Hubs

WGS84 GPS Coordinates

18

Tracked KPIs

Across 6 Urban Dimensions

12

Spatial GIS Algorithms

Pure-R Leaflet Cartography

ASSESSMENT TASK 01 (2 MARKS)

1. Frame the Problem

Case study context, target municipal audience, operational decision goals, dataset schema, and core analytical questions.

TASK 1.1 • PROBLEM CONTEXT

Vellore Smart City Urban Governance Challenge

Urban Fragmentation

Rapid expansion across 8 major sectors strains civic infrastructure and traffic management.

Siloed Metrics

Traffic, water, sanitation, and AQI data sit in isolated departmental spreadsheets.

Unit Incompatibility

Comparing vehicles/day against water supply (ML/d) requires mathematical standardization.

TASK 1.2 • STAKEHOLDERS & DECISIONS

Intended Audience & Supported Operational Decisions

Municipal Commissioners

Prioritize ward-level capital budget allocation and multi-sector infrastructure upgrades.

Traffic & Transit Planners

Reroute arterial corridors and optimize signal timings during peak congestion spikes.

Emergency Dispatch

Deploy maintenance units and rapid emergency response teams to highest-risk zones.

TASK 1.3 • DATASET ATTRIBUTES

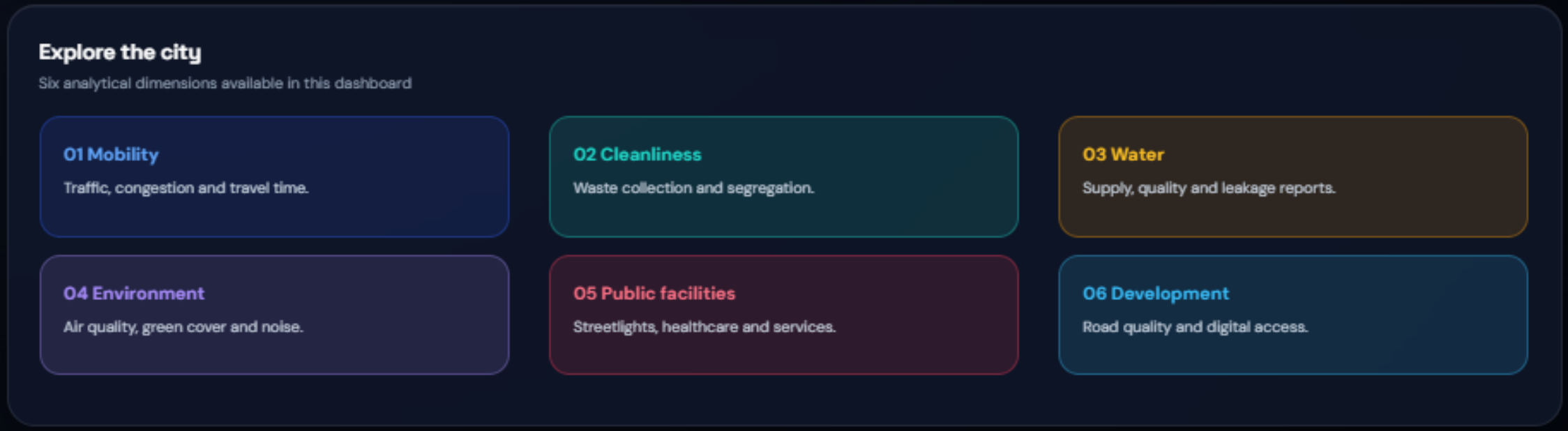
Data Schema & Measurement Scales in R

Attribute Name	R Data Type	Visualization Scale	Domain / Cardinality
dimension	character	Nominal (Categorical)	6 Urban Themes (Mobility, Water, etc.)
location	character	Nominal (Spatial Key)	8 Monitored Vellore Hubs
date	Date (YYYY-MM-DD)	Temporal (Interval)	21 Monthly Steps (Jan 2025 – Sep 2026)
indicator	character	Nominal (Sub-category)	18 Civic Metrics (3 per dimension)
value	numeric (double)	Quantitative (Ratio)	Continuous bounded range per metric
latitude / longitude	numeric (double)	Spatial (Geographic)	WGS84 Decimal Degrees

TASK 1.4 • INDICATOR CATALOGUE

The 6 Urban Dimensions & Bounded Value Ranges

Dimension	Indicator 1 (Range)	Indicator 2 (Range)	Indicator 3 (Range)
01 Mobility	Traffic Volume (1500–9500)	Travel Time (8–45 min)	Congestion (10–100%)
02 Cleanliness	Waste Collected (200–2000 t)	Segregation (10–95%)	Waste Issues (0–40)
03 Water	Water Supply (2–20 ML/d)	Water Quality (40–100)	Leakage Reports (0–30)
04 Environment	Air Quality Index (20–200)	Green Cover (5–50%)	Noise Level (35–95 dB)
05 Public Facilities	Streetlights (30–100%)	Healthcare Access (20–100)	Service Requests (0–60)
06 Development	Road Quality (20–100)	Construction (0–35 sites)	Digital Access (20–100%)



TASK 1.5 • ANALYTICAL FORMULATION

Two Core Analytical Questions for Vellore Leadership

ANALYTICAL QUESTION 1 (CROSS-SECTIONAL & MULTI-METRIC)

Which Vellore zones suffer severe multi-sector deficits across incompatible measurement units?

Justification: Enables commissioners to compare Traffic (vehicles/day) against Water Quality (score) simultaneously to identify systemic stress hotspots.

ANALYTICAL QUESTION 2 (SPATIO-TEMPORAL & TERRITORIAL)

How does Traffic Volume intensity partition across spatial catchments and evolve monthly?

Justification: Helps transit engineers define service boundaries around sensor hubs and detect seasonal bottlenecks across the 21-month timeline.

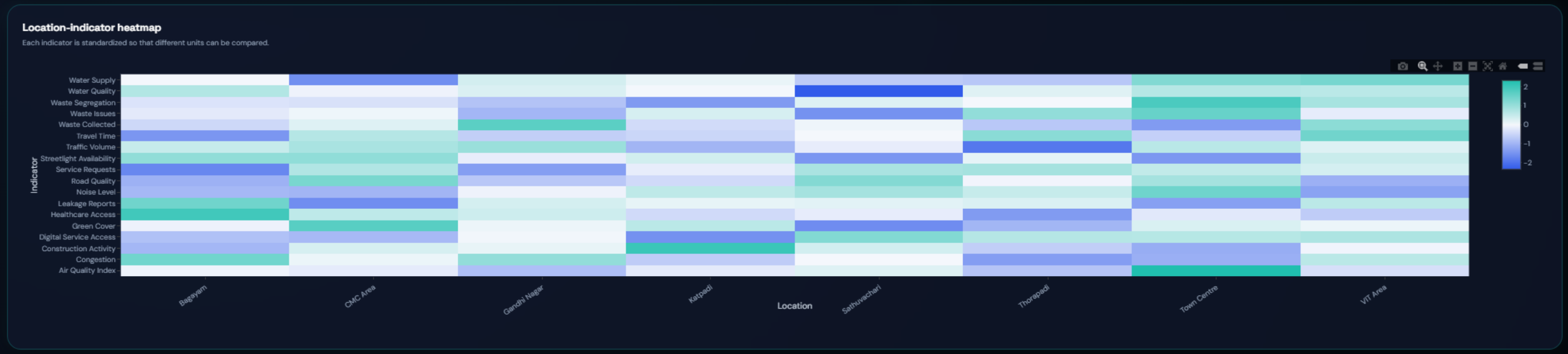
ASSESSMENT TASK 02 (3 MARKS)

2. Choose & Defend Visualizations

Visual encoding specifications, labelled architectural sketches, R implementation, and comparative defense against alternative techniques.

TASK 2.1 • PROPOSED VISUALIZATION 1 (QUESTION 1)

Labelled Sketch: Standardized Z-Score Matrix Heatmap



TASK 2.2 • TECHNIQUE 1 JUSTIFICATION

Why Z-Score Heatmap Beats Grouped Bar Charts

✓ Selected: Z-Score Matrix Heatmap

Zero Scale Distortion: Standardizes vehicles/day (9,500) and Water Supply (20 ML/d) onto [−2, +2].

Compact Matrix Density: Displays 8 × 18 = 144 cells without visual clutter.

Preattentive Hotspot Detection: Color saturation immediately flags anomaly cells.

✗ Alternative: 18-Series Grouped Bar Chart

Severe Y-Axis Compression: Traffic bars (9,000) crush Water Supply bars (15) to invisible 1px lines.

Visual Clutter: 144 adjacent bars exceed working memory limits.

Legend Overload: 18 distinct categorical hues become indistinguishable.

TASK 2.4 • TECHNIQUE 2 JUSTIFICATION

Why 2D Voronoi Catchments Beat 3D Extruded Columns

✓ Selected: 2D Voronoi Catchment Map

Zero Perspective Foreshortening: Planar 2D polygons preserve true geographic bounds.

Clear Territorial Partition: Every street in Vellore maps to its closest sensor node.

No Occlusion: Katpadi and VIT share a clean boundary wall.

✗ Alternative: 3D Extruded Column Map

Severe Occlusion: Tall southern columns hide northern Katpadi markers.

Perspective Tilt Distortion: Viewers cannot judge 3D bar heights accurately against tilted axes.

False Building Semantics: Extruded pillars look like skyscrapers rather than data.

ASSESSMENT TASK 03 (3 MARKS)

3. Critique & Correct

Deliberately flawed visualization sketch, identification of two perceptual/encoding errors, decision impact analysis, and corrected visual design.

TASK 3.1 • IMPACT ON MUNICIPAL DECISIONS

How Visual Encoding Errors Mislead Urban Policy

Lie Factor Distortion

Truncating the bar baseline breaks length-proportional perception. Officials assume an existential 8× crisis in Katpadi and overlook systemic citywide transit trends.

Semantic Stroop Interference

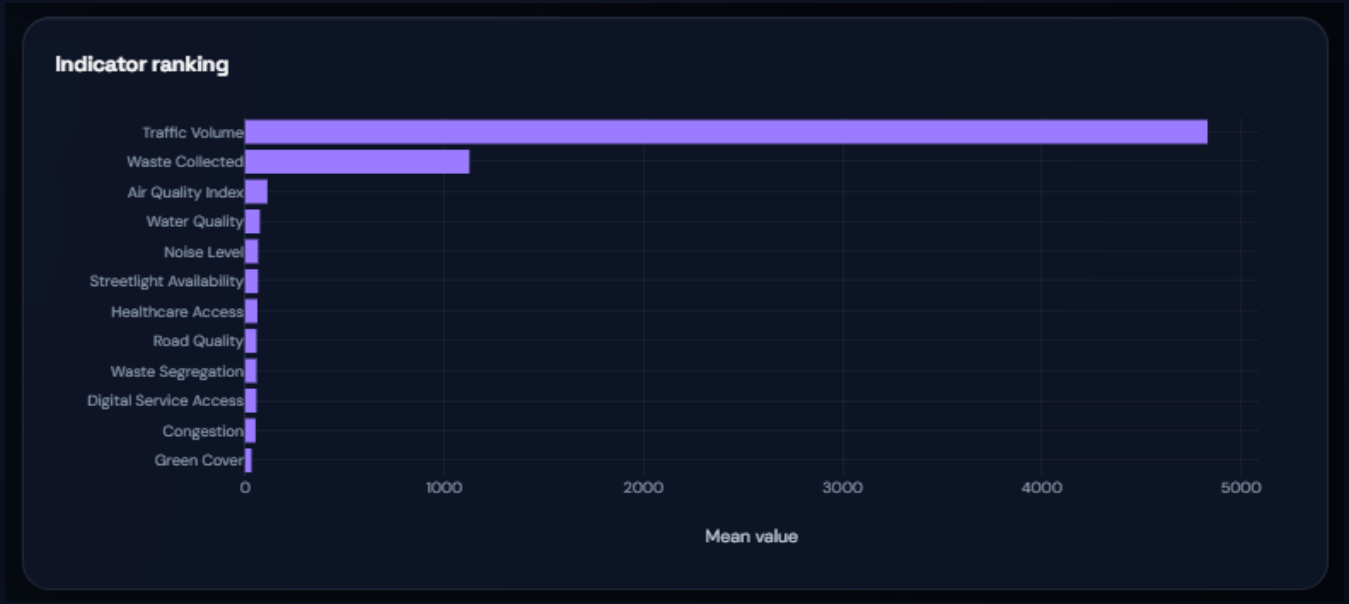
Green universally signals safety. Executives glancing at dashboards misread 66% severe gridlock as acceptable and healthy operating conditions.

Misallocated Patrol Units

Traffic police squads get dispatched to Bagayam due to deceptive red alert hues, leaving genuine Katpadi bottlenecks completely unmanned.

TASK 3.2 • CORRECTED VISUALIZATION SKETCH

Corrected Design: Zero-Baseline Ranked Horizontal Bar Chart



Revised Design Justification

Lie Factor = 1.0: Full 0–100% axis ensures bar length equals exact numerical ratio.

Sequential Cyan-Blue: Avoids red-green deuteranopia confusion and Stroop traps.

Horizontal Labels: Eliminates -35° neck-tilting text rotation for effortless scanning.

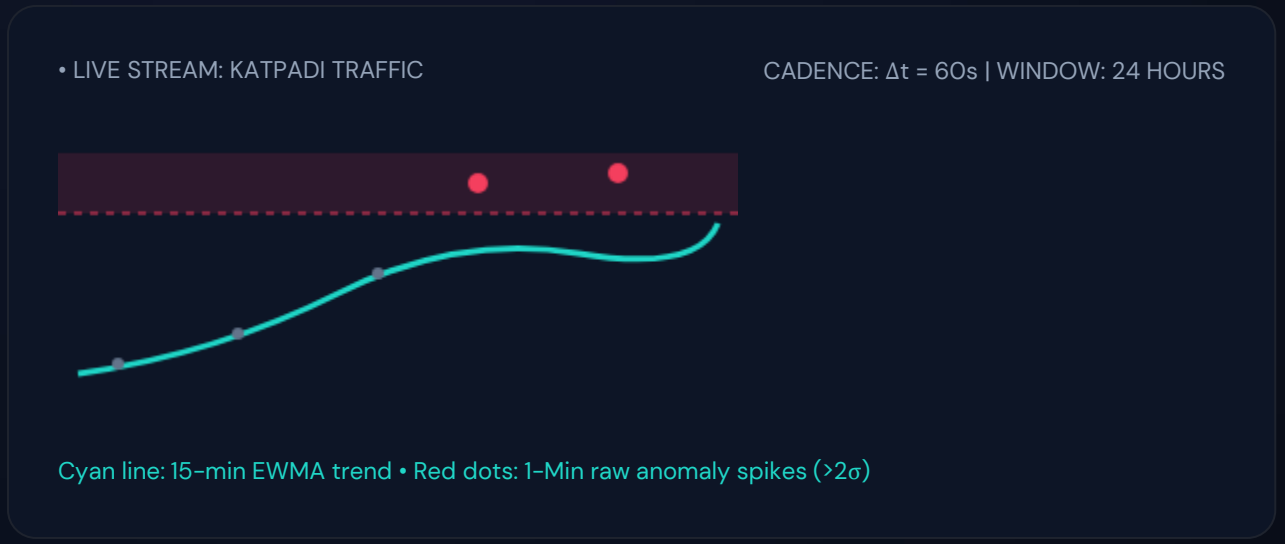
ASSESSMENT TASK 04 (2 MARKS)

4. Respond to a Changed Requirement

Adapting static monthly visualizations for continuous streaming IoT telemetry, update cadence, sliding windows, and smoothing trade-offs.

TASK 4.1 • STREAMING VISUALIZATION ADAPTATION

Dual-Layer Horizon Strip: Raw Dots + 15-Min EWMA



Streaming Parameters

Update Interval: 60-second micro-batches via reactiveTimer(60000).

Display Window: 24-hour FIFO sliding window (1,440 ticks) + 60-min zoom inset.

Dual Encoding: Faint scatter dots preserve raw spikes; bold line shows 15-min EWMA trend.

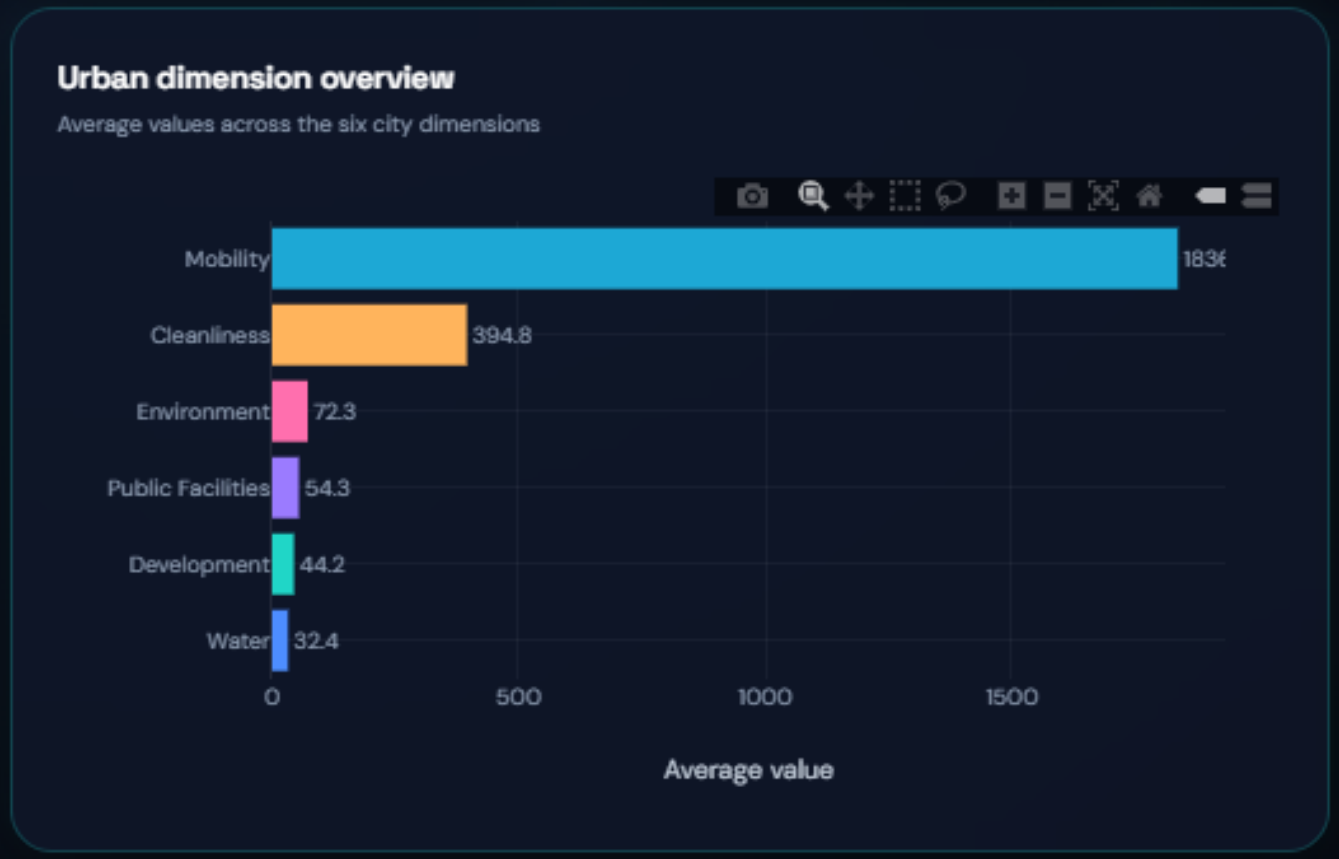
TASK 4.2 • STREAMING DESIGN JUSTIFICATION

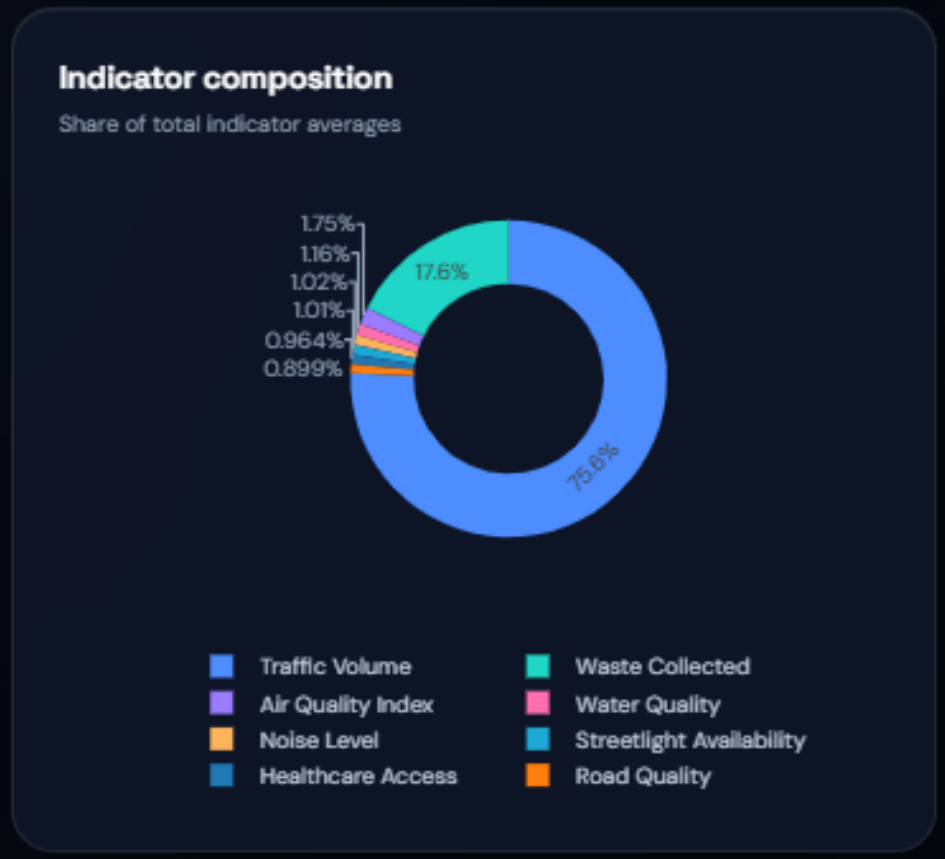
Trade-Off: Detecting Sudden Spikes vs. Revealing Trends

Design Parameter	Raw Unaggregated Stream	Heavy Hourly Aggregation	Proposed Dual-Layer Hybrid
Sudden Spike Detection	Instantaneous (<1 min)	Delayed / Smoothed away	**Instant via 2σ anomaly dots**
Overall Diurnal Trend	Obscured by high-frequency noise	Clear macro shape	**Clear via 15-min EWMA curve**
Cognitive Load	High visual jitter & false alarms	Low, but misses emergencies	**Balanced preattentive hierarchy**
R Shiny Implementation	Full DOM redraw every second	Static hourly poll	**plotlyProxy(extendTraces) at 60s**

R CODE ANALYSIS • HOME DASHBOARD

Dimension Bar (`home_dimensions`) & Top-8 Donut





PART III • COMPUTATIONAL GEOMETRY & GIS ENGINE

5. Pure-R Geospatial Algorithms

Analysis of the zero-dependency spatial geometry functions and the 12 interactive cartographic visualization modes.

R CODE ANALYSIS • LINES 2549–2668

Geospatial Engine: Spatial Methods 01 to 06

Method ID & Name	Leaflet Geometry Primitive	Mathematical Scaling / Logic
01 Point Map (Radar Glow)	addCircleMarkers + addCircles	Fixed 9px core + 550m dashed radar ring
02 Proportional Symbols	Dual addCircles + addCircleMarkers	Flannery area scaling: 7 + 20 * sqrt(scaled)
03 Cluster Map	markerClusterOptions()	Dynamic zoom-dependent spatial aggregation
04 Choropleth (Voronoi)	addPolygons via compute_voronoi_cells	Nearest-neighbor half-plane territorial partitioning
05 Cartogram	3 Concentric graduated addCircles	Value-distorted rings: c(350, 750, 1300) * (0.6 + 1.2*s)
06 Hex Binning	addPolygons via make_hex_grid_pure	IDW interpolation across regular hexagonal tessellation

R CODE ANALYSIS • LINES 2670–2847

Geospatial Engine: Spatial Methods 07 to 12

Method ID & Name	Leaflet Geometry Primitive	Mathematical Scaling / Logic
07 Heat Map (Multi-Tier)	4 Concentric Gaussian addCircles	1600m (#315BEA), 1000m (#20D6C7), 550m (#FFB45C), 220m (#D7191C)
08 Topographic Contours	5 Hypsometric addCircles bands	Elevation proxy bands (1400m down to 160m)
09 Flow Map (Bezier Arcs)	addPolylines via make_bezier_arc	Connects rank-ordered hubs (#1 → #2 → … → #8)
10 Spider Map (Starburst)	Radial addPolylines from which.max	Hub-to-spoke rays from peak node to all 7 peers
11 Time-Space Trajectory	Chronological addPolylines	Links sequential temporal observations across space
12 Data Space Proximity	Euclidean dist(scale(mat)) graph	Connects each hub to its 2 most similar multi-KPI peers

SPATIAL TECHNIQUE 01 DETAIL • LINES 2549–2565

01 Point Map (Radar Glow)



Geometry & Cartographic Architecture

Leaflet Geometry Primitive: Dual addCircleMarkers (fixed 9px solid white core) and addCircles (550m radius dashed radar ring).

Mathematical Scaling: Core marker radius is invariant to map zoom level; dashed perimeter circle retains real-world 550m geographic metric bounds.

Visual Encodings: 4-tier numeric palette maps continuous KPI values directly to marker fills with frosted glass tooltips.

Operational Use Case: Pinpoints discrete municipal sensor nodes and flags localized threshold breaches without obscuring the street grid.

SPATIAL TECHNIQUE 02 DETAIL • LINES 2567–2582

02 Proportional Symbols



Flannery Area Compensation

Leaflet Geometry Primitive: Dual addCircles (concentric catchment halo) and addCircleMarkers (proportional disk).

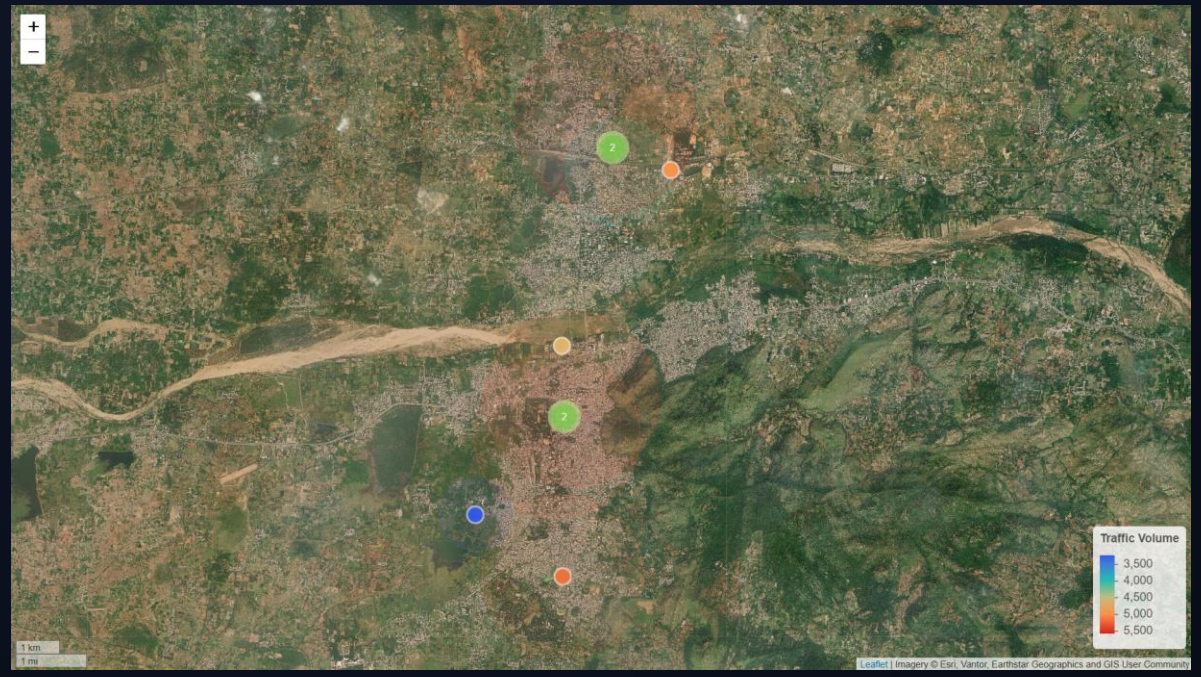
Mathematical Scaling: Square-root area scaling formula: 7 + 20 * sqrt(scaled), correcting for human visual underestimation of large circular areas.

Visual Encodings: Symbol area directly reflects metric volume while halo rings expand to (400 + 1400 * sqrt(scaled)) meters.

Operational Use Case: Enables instantaneous ratio comparison of traffic volume and solid waste mass between disparate wards.

SPATIAL TECHNIQUE 03 DETAIL • LINES 2584–2592

03 Cluster Map



Density Aggregation & De-cluttering

Leaflet Geometry Primitive: markerClusterOptions() integrated on Leaflet marker layers.

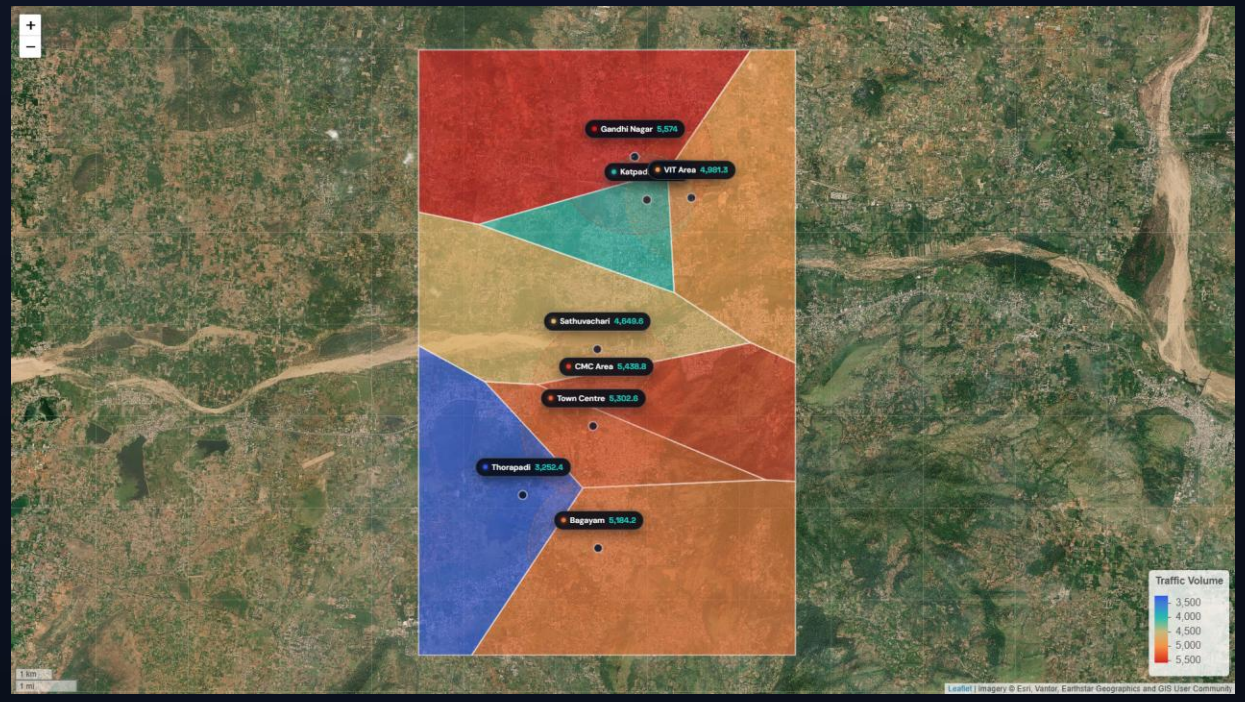
Mathematical Scaling: Dynamic screen-space distance clustering that hierarchically coalesces adjacent markers based on current viewport zoom levels.

Visual Encodings: Clustered badges display total aggregated point counts; clicking triggers a animated spiderfy fan-out.

Operational Use Case: Eliminates symbol collision and visual fatigue when scaling up to thousands of citizen service tickets.

SPATIAL TECHNIQUE 04 DETAIL • LINES 2594–2618

04 Choropleth (Voronoi)



Geometric Catchment Partitioning

Leaflet Geometry Primitive: addPolygons rendered via pure-R compute_voronoi_cells function.

Mathematical Scaling: Sutherland-Hodgman half-plane clipping: partitions 2D space into regions where every point is closest to its seed hub: $V(p_i) = \{x \mid \|x - p_i\| \le \|x - p_j\|\}$.

Visual Encodings: Polygon fills colored by indicator intensity (62% opacity) with #20D6C7 highlight borders on hover.

Operational Use Case: Partitions municipal service zones and attributes urban space to nearest maintenance hubs without official boundary shapefiles.

SPATIAL TECHNIQUE 05 DETAIL • LINES 2620–2636

05 Cartogram



Value-Distorted Spatial Geometry

Leaflet Geometry Primitive: 3 concentric graduated addCircles + centered addCircleMarkers.

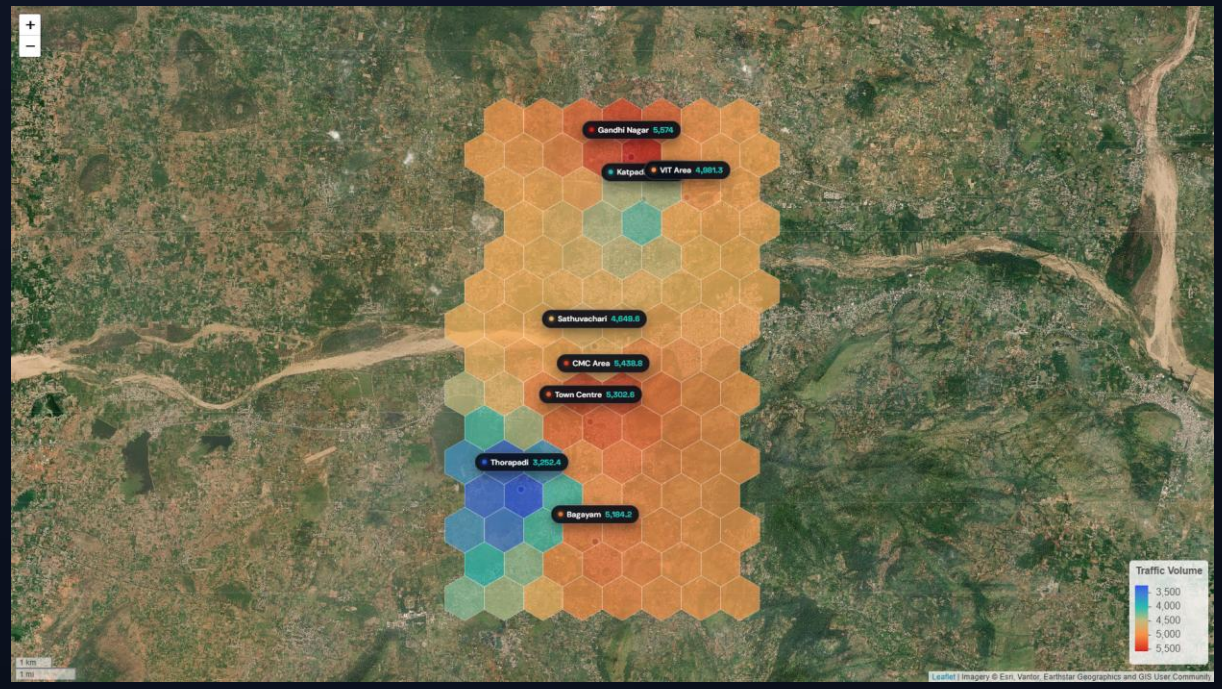
Mathematical Scaling: Value distortion rings scaled by: c(350, 750, 1300) * (0.6 + 1.2 * scaled) with inner core (8 + 16 * scaled) px.

Visual Encodings: Expands visual territory based on data severity rather than physical geographical footprint.

Operational Use Case: Forces executive focus onto severe urban congestion zones regardless of the small geographic footprint of central wards.

SPATIAL TECHNIQUE 06 DETAIL • LINES 2638–2668

06 Hex Binning



Adaptive Mesh Tessellation

Leaflet Geometry Primitive: Regular hexagonal addPolygons computed via make_hex_grid_pure.

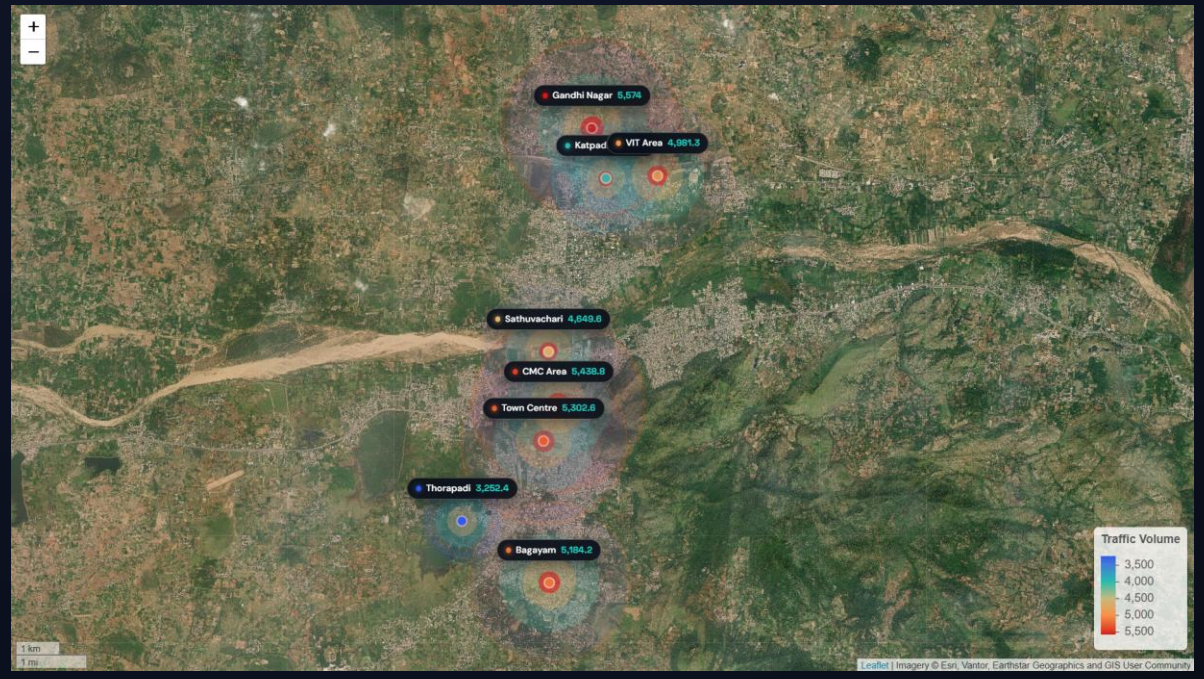
Mathematical Scaling: Cosine-corrected grid steps: $dx = \sqrt{3} \cdot r_{\text{lng}}, dy = 1.5 \cdot r_{\text{lat}}$; IDW weighting: $w_i = 1 / (d_i^2)^{1.5}$.

Visual Encodings: Dynamic zoom observer rescales hexagon radii from 1,800m (Zoom ≤11) down to 180m (Zoom ≥15).

Operational Use Case: Interpolates environmental AQI values continuously across regions between sparse monitoring sensors.

SPATIAL TECHNIQUE 07 DETAIL • LINES 2670–2696

07 Heat Map (Multi-Tier)



Concentric Gaussian Gradient

Leaflet Geometry Primitive: 4 concentric Gaussian addCircles per node + center marker.

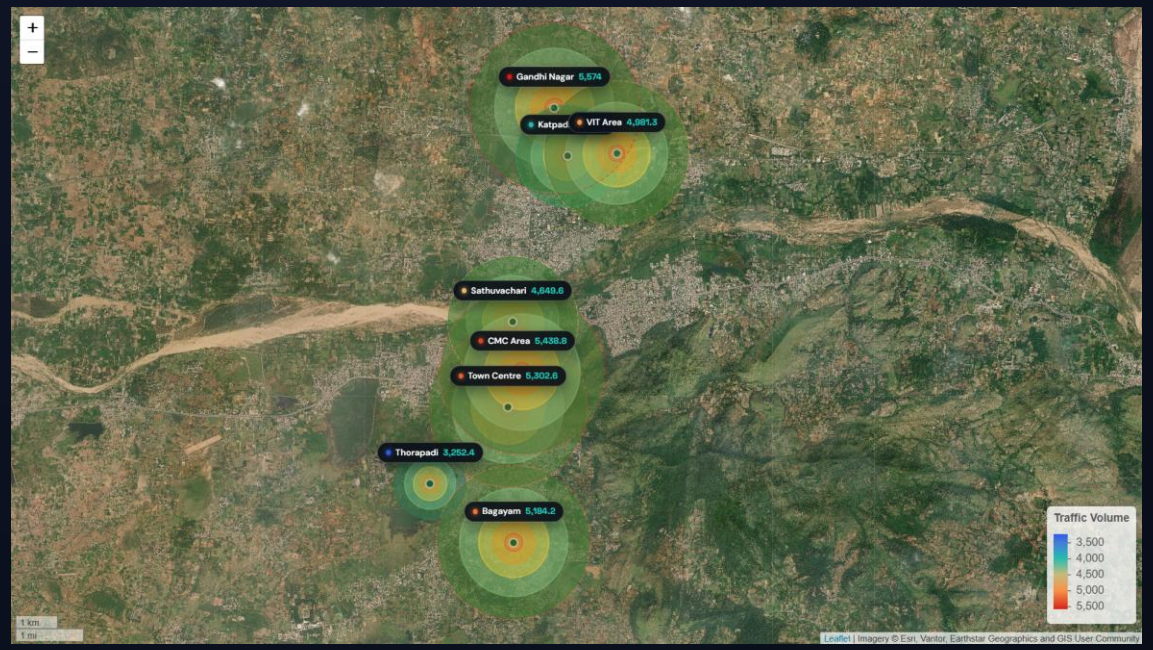
Mathematical Scaling: Tiered radial kernels: 1600m ambient (opacity 0.08), 1000m intermediate (0.16), 550m acute (0.28), and 220m core (0.65).

Visual Encodings: Monotonically increasing color temperature from deep navy through cyan and amber to red.

Operational Use Case: Rapid preattentive detection of acute traffic gridlock and noise pollution epicenters.

SPATIAL TECHNIQUE 08 DETAIL • LINES 2698–2718

08 Topographic Contours



Hypsometric Elevation Tiers

Leaflet Geometry Primitive: 5 concentric hypsometric addCircles bands per location.

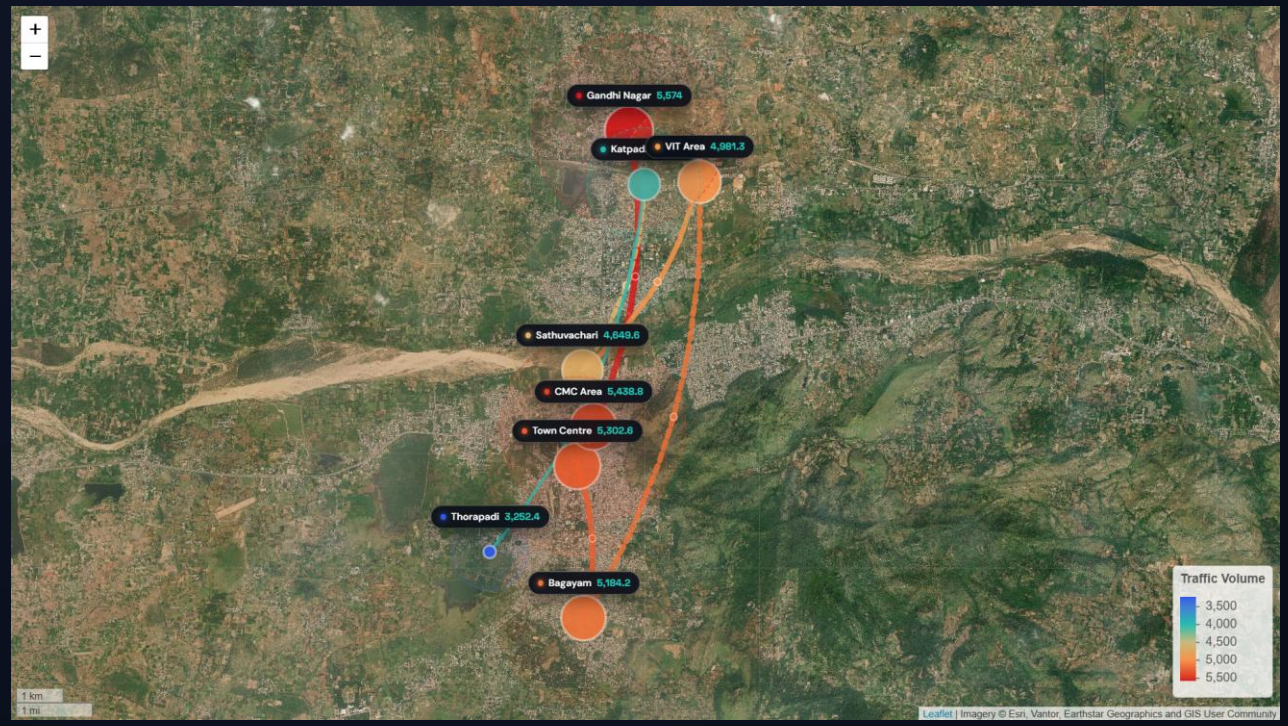
Mathematical Scaling: Elevation proxy bands: radii at c(1400, 1000, 650, 380, 160) * (0.6 + 0.8 * scaled) meters.

Visual Encodings: Classical topographic palette: #2E7D32 (green base), #81C784, #FDD835, #FB8C00, #E53935 (peak red).

Operational Use Case: Depicts intensity gradients as continuous isolines, aiding in drainage and contaminant runoff studies.

SPATIAL TECHNIQUE 09 DETAIL • LINES 2720–2752

09 Flow Map (Bezier Arcs)



Curved Directional Corridors

Leaflet Geometry Primitive: addPolylines computed via parametric make_bezier_arc.

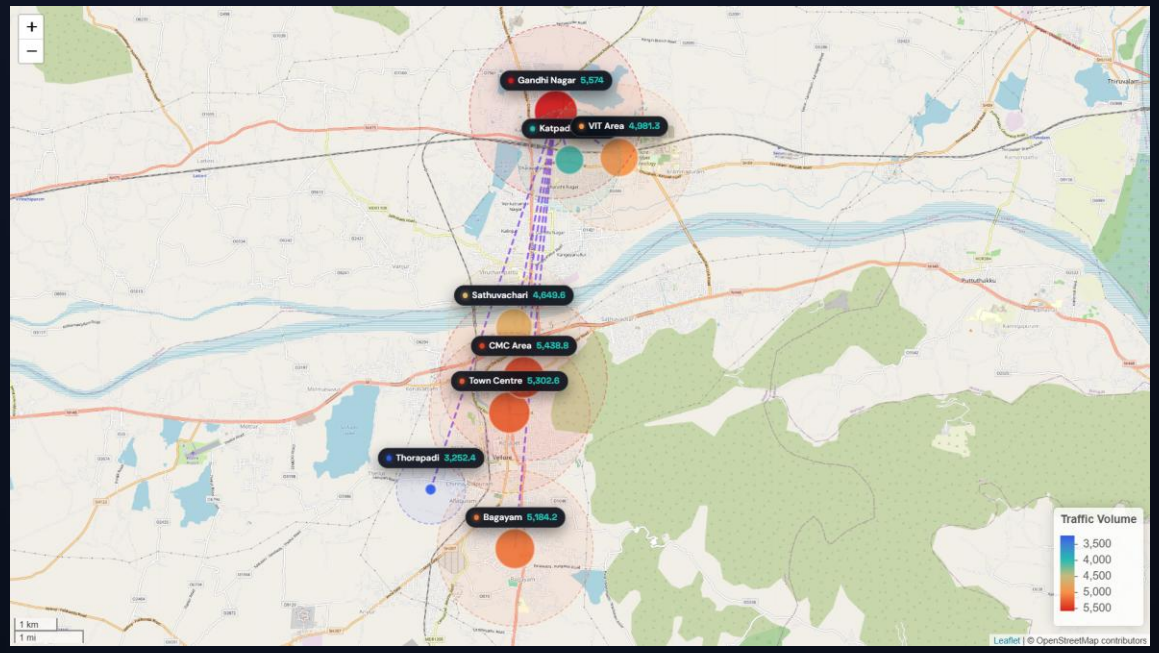
Mathematical Scaling: Quadratic Bezier formula with 28 interpolation ticks: $B(t) = (1-t)^2 P_0 + 2(1-t)t P_c + t^2 P_1$ with curvature ± 0.18.

Visual Encodings: Stroke width scaled by pmax(2.5, 2 + 5 * scaled); dashed styling on alternating paths.

Operational Use Case: Visualizes inter-district mobility flow volumes along ranked urban arterial transit corridors.

SPATIAL TECHNIQUE 10 DETAIL • LINES 2754–2784

10 Spider Map (Starburst)



Radial Hub-and-Spoke Vectors

Leaflet Geometry Primitive: Radial addPolylines from which.max hub + range circles.

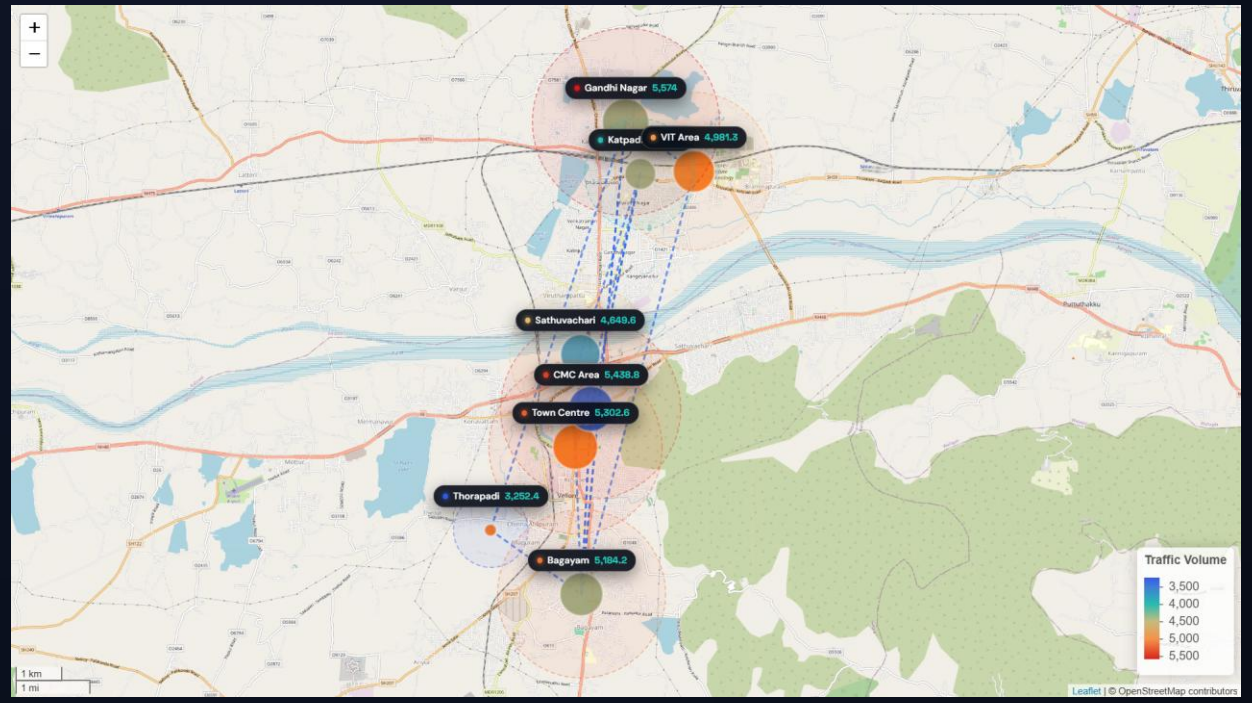
Mathematical Scaling: Haversine geodesic ray distance calculation from active epicenter hub to all 7 peripheral nodes.

Visual Encodings: #FFB45C dashed concentric boundary range rings (400m, 900m, 1500m) with #7C3AED violet connecting rays.

Operational Use Case: Pinpoints dependence of peripheral zones on the primary urban center for healthcare or university transit.

SPATIAL TECHNIQUE 11 DETAIL • LINES 2786–2814

11 Time-Space Trajectory



Spatio-Temporal Sequence Paths

Leaflet Geometry Primitive: Chronological addPolylines linking sequential observations.

Mathematical Scaling: Time-ordered coordinate array: connects temporal peak shifts $(t_1 \to t_2 \to \dots \to t_{21})$ across 21 monthly observations.

Visual Encodings: Continuous chromatic temporal ramp (#2563EB early months $\to$ #14B8A6 mid-stage $\to$ #F97316 recent months).

Operational Use Case: Tracks temporal migration of seasonal water scarcity and peak traffic congestion across Vellore.

SPATIAL TECHNIQUE 12 DETAIL • LINES 2816–2847

12 Data Space Proximity



Multivariate Similarity Graph

Leaflet Geometry Primitive: Multi-attribute topological addPolylines linking similar peers.

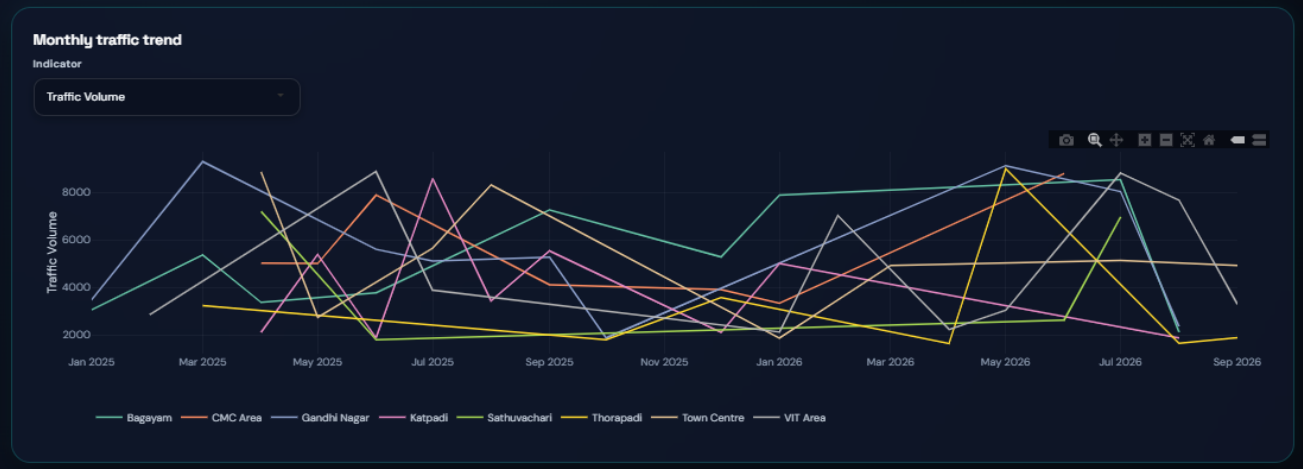
Mathematical Scaling: Euclidean distance matrix in standardized 18-KPI feature space: dist(scale(mat)), connecting each node to its $k=2$ nearest data peers.

Visual Encodings: #0F766E dashed similarity cords with popup reporting multi-indicator profile match percentage.

Operational Use Case: Discovers non-contiguous urban hubs sharing identical civic distress patterns for unified policy intervention.

R CODE ANALYSIS • LINES 3001–3140

Domain Tabs: Multi-Line Trends & Boxplot Distributions

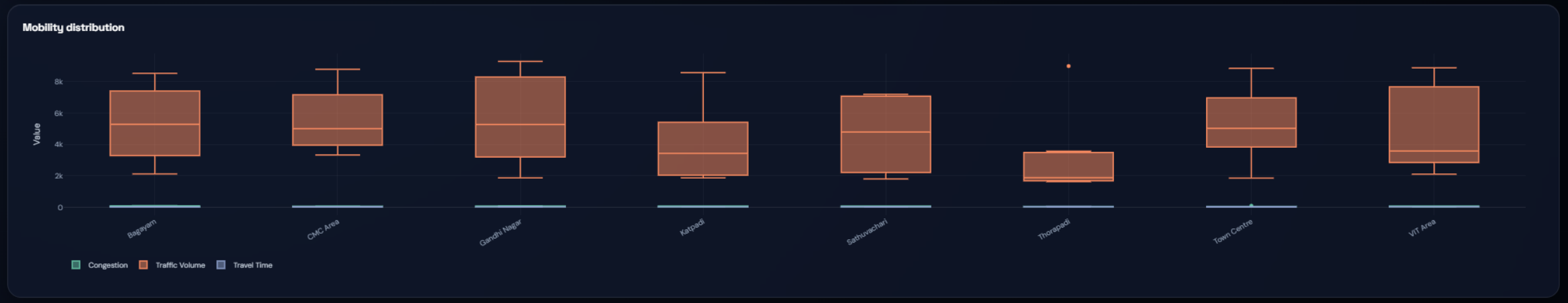


Statistical Encodings

Non-Parametric Spread: Boxplots reveal median, interquartile range (IQR), skewness, and extreme outlier months.

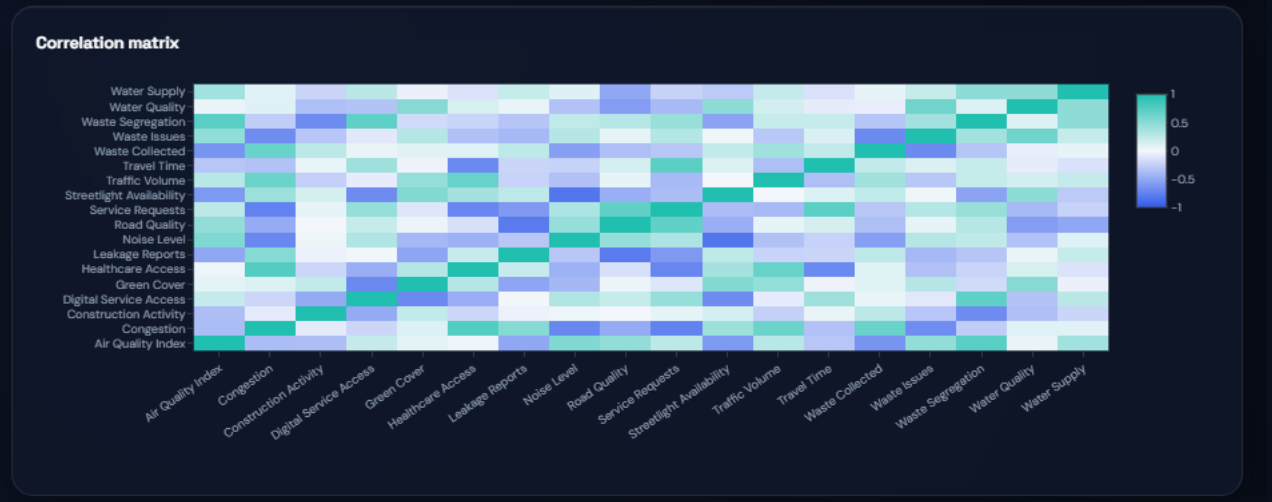
Combined Environment & Water: Filters d$dimension %in% c("Environment", "Water") for cross-theme comparison.

Interactive Outliers: Points plotted via boxpoints = "outliers" highlight acute sensor failures.



R CODE ANALYSIS • LINES 3142–3284

Visual Analytics Lab: Pairwise Correlation & Scatter





Multivariate Laboratory

Pairwise Complete Correlation: cor(wide, use = "pairwise.complete.obs") bounded to zmin = −1, zmax = 1.

Overlay Histograms: barmode = "overlay", nbinsx = 22, opacity = 0.55 compares distribution density across locations.

Bivariate Associations: Inspects relationships like Traffic Volume vs. Ambient Noise.

R CODE ANALYSIS • LINES 1720–1940

7-Stage CSV Validation & Nominatim Geocoding Pipeline

STAGE 01

Schema & Dates

Verifies 6 required columns & strict YYYY-MM-DD parsing.

STAGE 02

Deduplication

Removes natural-key duplicates & verifies unit consistency.

STAGE 03

Plausibility

Flags out-of-range numbers & calendar day gaps.

STAGE 04

OSM Geocoding

Resolves unknown place names via OpenStreetMap Nominatim.

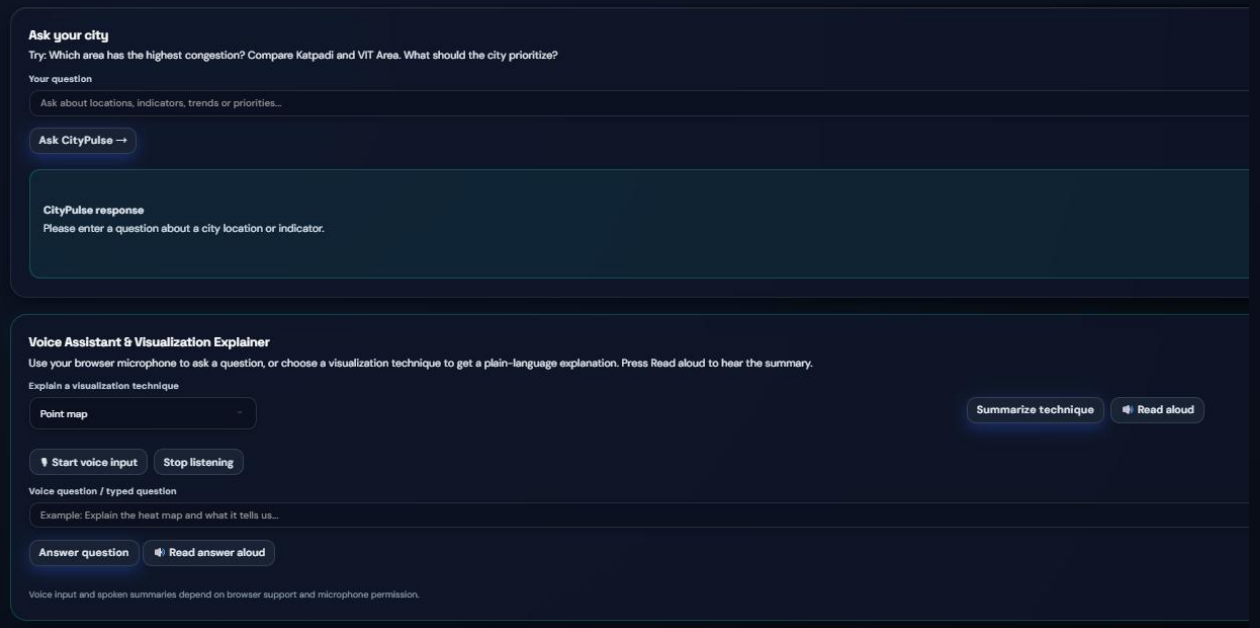
PART IV • DECISION INTELLIGENCE & AI MODULES

6. AI Assistant, Simulation & Routing

Rule-based NLP query engine, Web Speech API integration, what-if forecasting models, 3D Digital Twin, Resource Optimizer, and OSRM road routing.

R CODE ANALYSIS • LINES 664–736 & 3315–3468

Rule-Based NLP Assistant & Browser Voice Synthesis



Assistant Architecture

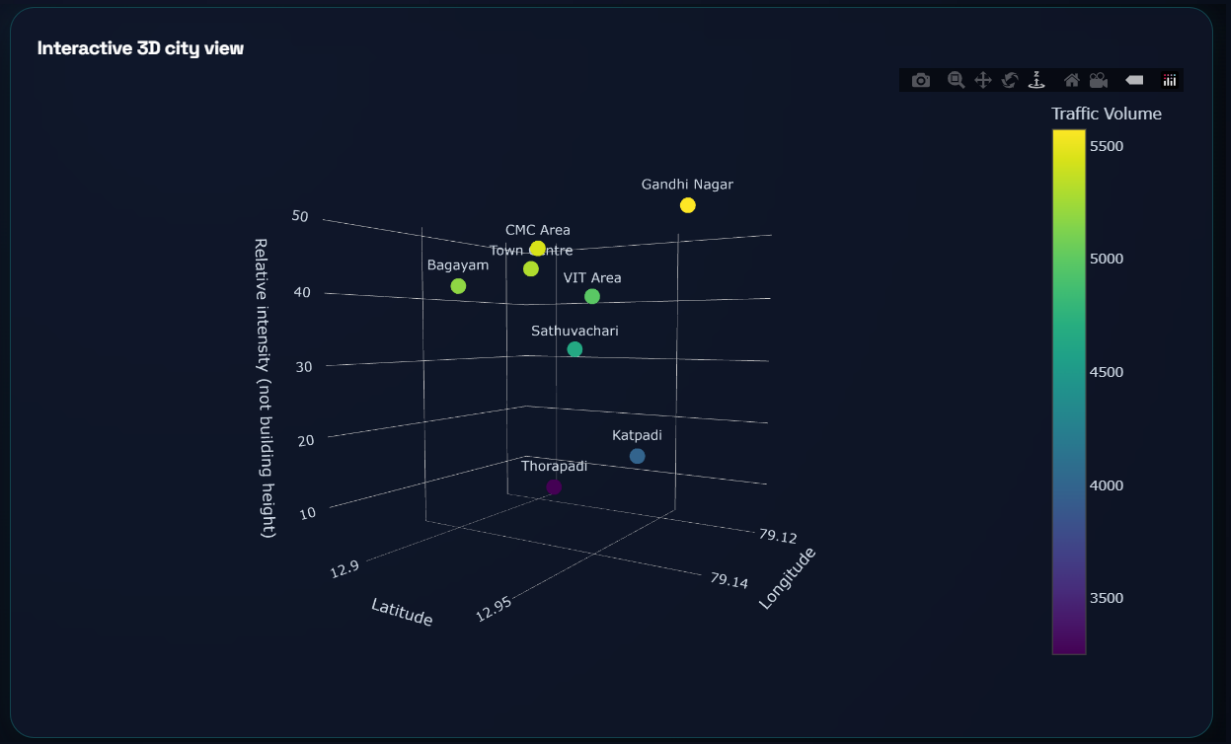
Zero Hallucination Guarantee: Uses transparent regex intent matching (compare|versus, highest|worst, priorit|risk) over active data.

Browser SpeechRecognition (`en-IN`): Captures spoken Indian-English queries.

SpeechSynthesis Playback: Reads answers and 18 visualization technique explainers aloud at rate = 0.95.

R CODE ANALYSIS • LINES 3800–3838

3D Digital Twin: WebGL `scatter3d` Intensity Mapping



WebGL 3D Mechanics

Normalized Z-Elevation: Maps indicator intensity onto $z \in [5, 50]$ over WGS84 GPS coordinates.

Perceptually Uniform Viridis: Encodes value magnitude redundantly on both Z-height and marker color.

Ethical Disclaimer: Explicitly labels Z-axis as *"Relative intensity (not building height)"*.

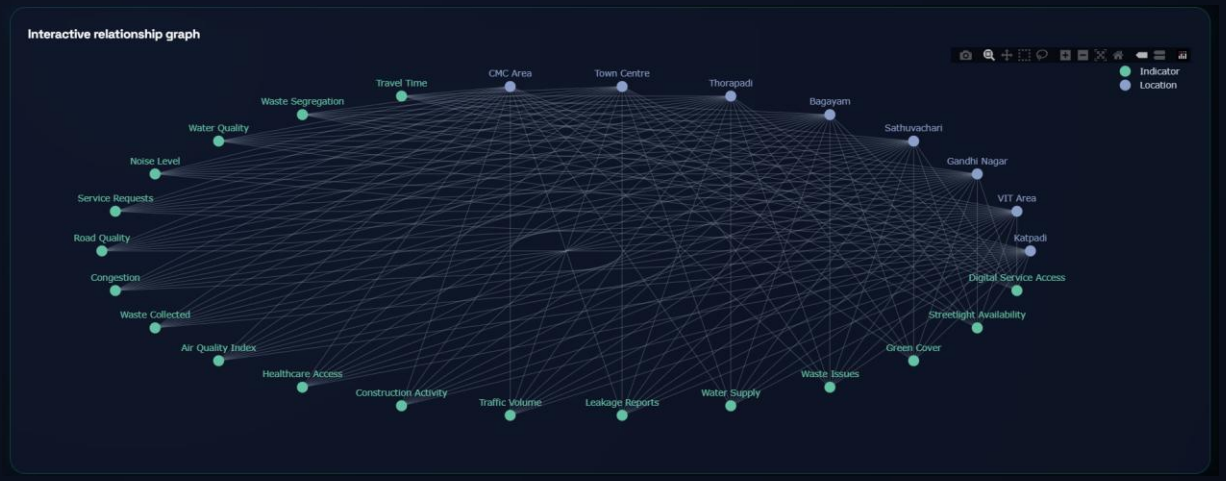
R CODE ANALYSIS • LINES 3840–3894

Resource Optimizer: Polarity Inversion & 3 Heuristics

Allocation Strategy	Mathematical Rule in R	Operational Use Case
Polarity Normalization	if (ind %in% positive_inds) score <- 100 - score	Inverts Water/Road Quality so low scores = high need
O1 Highest Risk First	Orders descending by need_score; assigns 1 unit to top-K	Emergency response to acute hotspots
O2 Need Proportional	Hamilton largest-remainder: floor(units * w / sum(w))	Fair fleet distribution weighted by demand
O3 Balanced Coverage	Integer quotient units %/% nrow(agg) + top remainder	Routine citywide preventive patrol

R CODE ANALYSIS • LINES 3896–3944

Urban Knowledge Graph: Circular Bipartite Layout



Bipartite Graph Architecture

Trigonometric Node Placement: Positions n nodes evenly around the unit circle via $x = \cos(\theta), y = \sin(\theta)$.

Bipartite Edge List: Links Location nodes (#315BEA) to Indicator/Dimension nodes (#20D6C7).

Causality Guardrail: UI explicitly warns that graph edges represent schema relationships, not causal proof.

R CODE ANALYSIS • LINES 3470–3676

Smart Route Analysis: OSRM API with Geodesic Fallback

STEP 01

Query OSRM API

Requests GeoJSON alternatives with a strict 5-second socket timeout.

STEP 02

Priority Selection

Picks which.min(duration) or which.min(distance).

STEP 03

Offline Fallback

Computes Haversine straight line if OSRM is unreachable.

R ARCHITECTURE • SCIENTIFIC RIGOR

Visualization Ethics & Scientific Rigor

Synthetic Provenance Tag

Every generated record carries source = "Synthetic demonstration data", preventing synthetic demo data from being mistaken for real-world statistics.

Non-Causal Disclaimers

Knowledge Graph and Scatter plots display prominent disclaimers warning users that statistical correlation does not establish causal proof.

Strict Upload Rejections

Halts CSV ingestion immediately on missing columns or invalid dates rather than silently omitting corrupted records.

ASSESSMENT VERIFICATION

CSI3005 Rubric Completion & Evidence Matrix

Assessment Task	Weight	CityPulse 360° Report Deliverable	Covered Slides
1. Frame the Problem	2 Marks	Vellore context, 3 audiences, 1,008-row schema, Q1 & Q2 formulated	**Slides 4–10**
2. Choose & Defend	3 Marks	Z-Score Heatmap vs Grouped Bar; Voronoi vs 3D Extruded Columns	**Slides 11–15**
3. Critique & Correct	3 Marks	50%-truncated Y-axis + inverted green/red fixed via 0–100% horizontal bar	**Slides 16–19**
4. Streaming Adaptation	2 Marks	60s cadence, 24h FIFO window, raw 2σ anomaly dots + 15m EWMA curve	**Slides 20–22**
R Code Architecture & Spatial Engine	Lab Link	Audit of 12 GIS algorithms, CSV validator, AI Lab & OSRM routing	**Slides 23–43**

CSI3005 ADVANCED DATA VISUALIZATION TECHNIQUES • FALL SEM 2026–2027

Thank You & Questions

CityPulse 360° Urban Intelligence Platform — Manthena Sri Adithya (24MID0177) • Vellore, Tamil Nadu

Slot: E1 + TE1 (Theory) • Comprehensive Case Study Assessment & Spatial Visualization Architecture